Setup Airflow Environment (MWAA) on AWS

Summary of the Lab

In this lab, an AWS Managed Apache Airflow (MWAA) environment is created to orchestrate workflows on AWS. Since Airflow is not a serverless service, a managed environment is configured with supporting infrastructure such as S3, VPC, security groups, and IAM roles. An S3 bucket is used to store DAGs and dependency files, and a requirements.txt file is provided to install required Python libraries during environment creation. The environment class, worker scaling, and web server accessibility are configured before creating the Airflow environment.

Prerequisite for the Lab

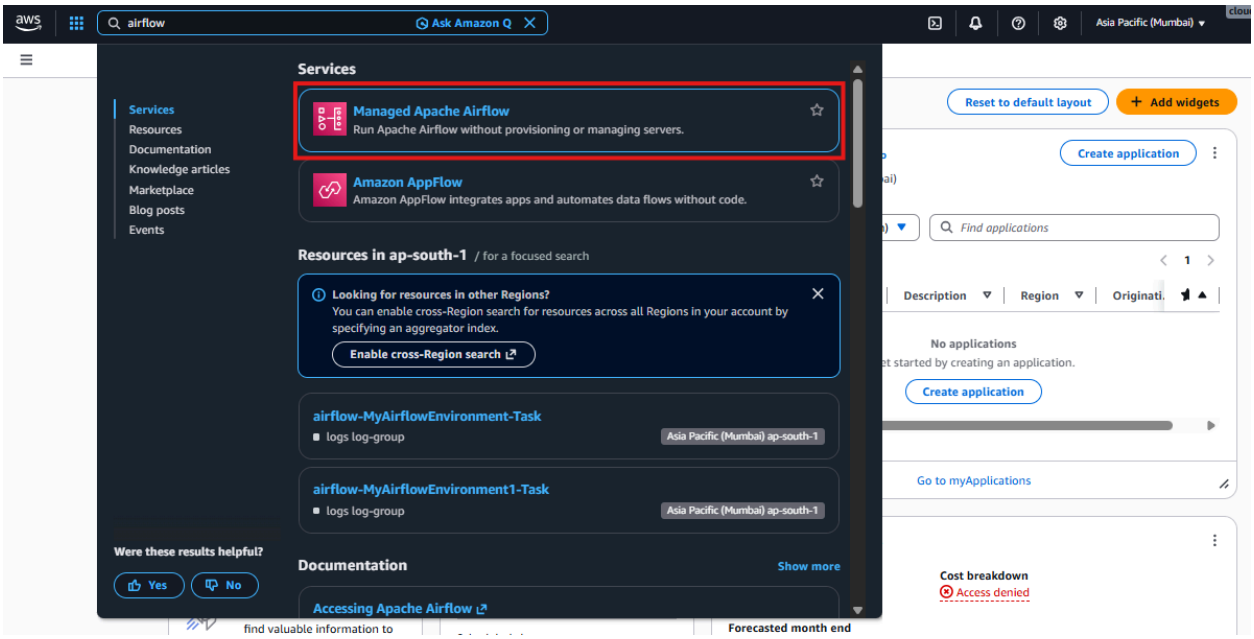
o AWS account with access to Managed Apache Airflow

o S3 bucket created for Airflow DAGs

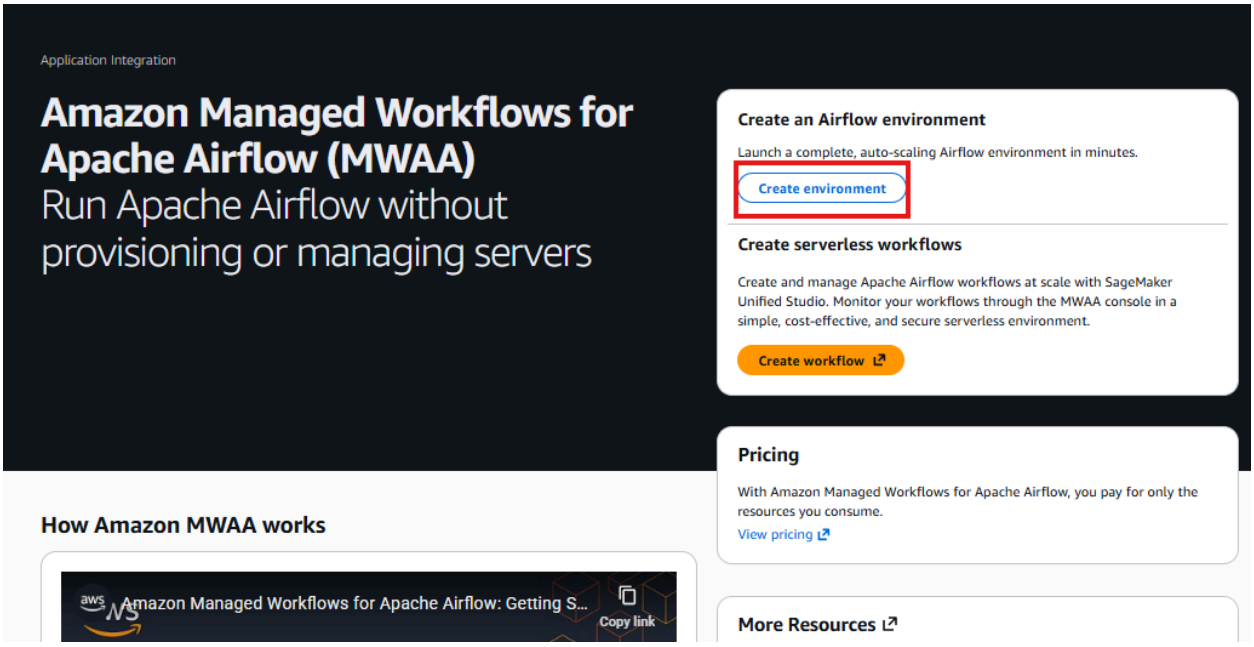
o requirements.txt file containing the pandas library

Steps to complete the lab

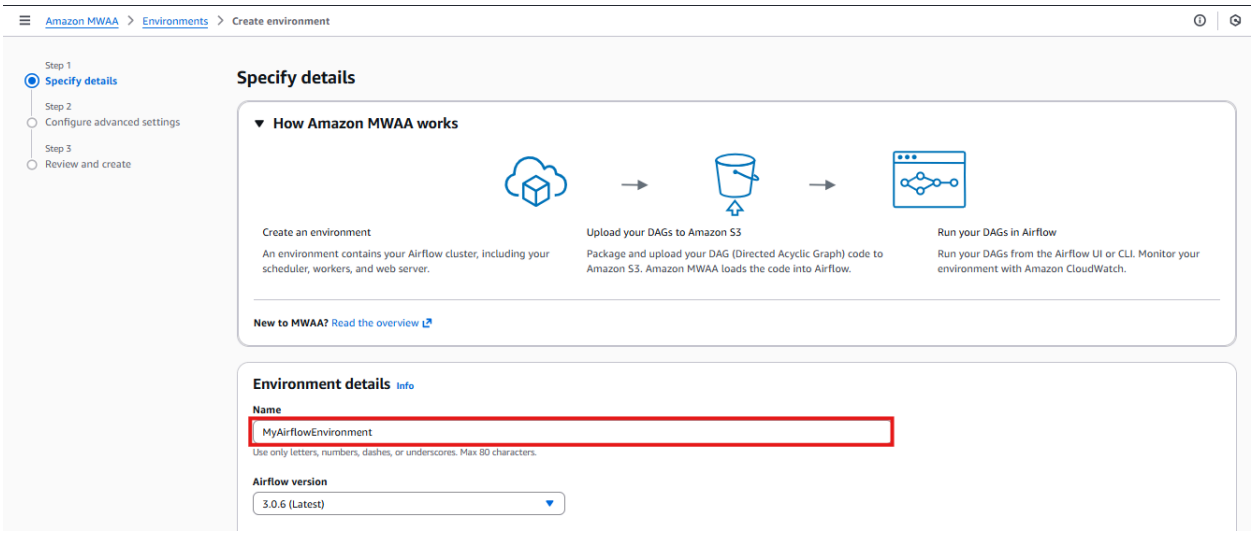
- Go to AWS Console → Managed Workflows for Apache Airflow (MWAA)

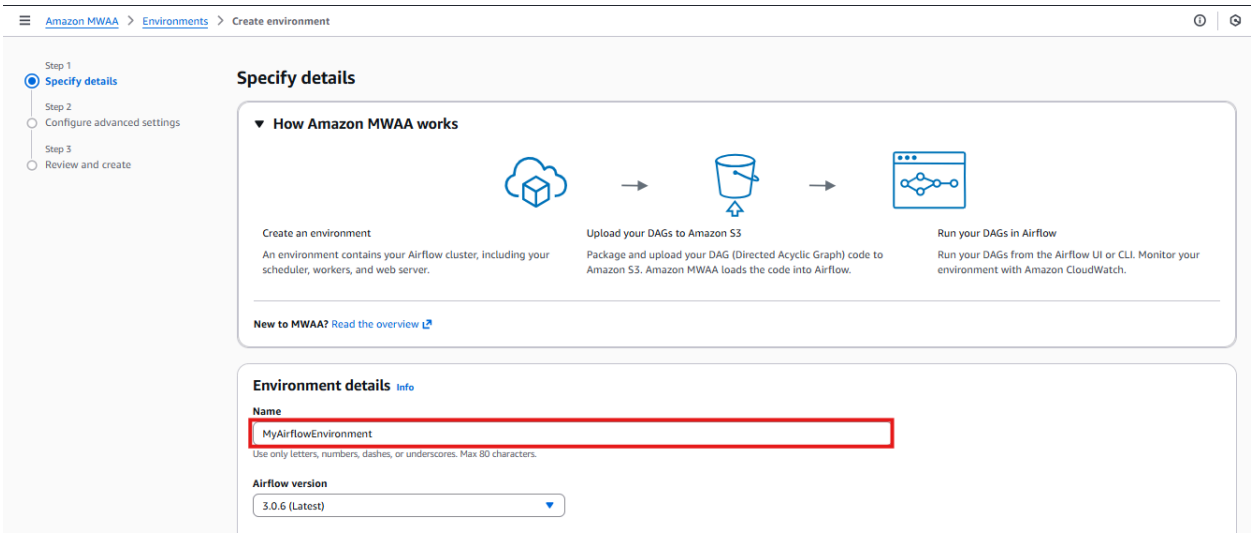


- Click on My environments → Create environment

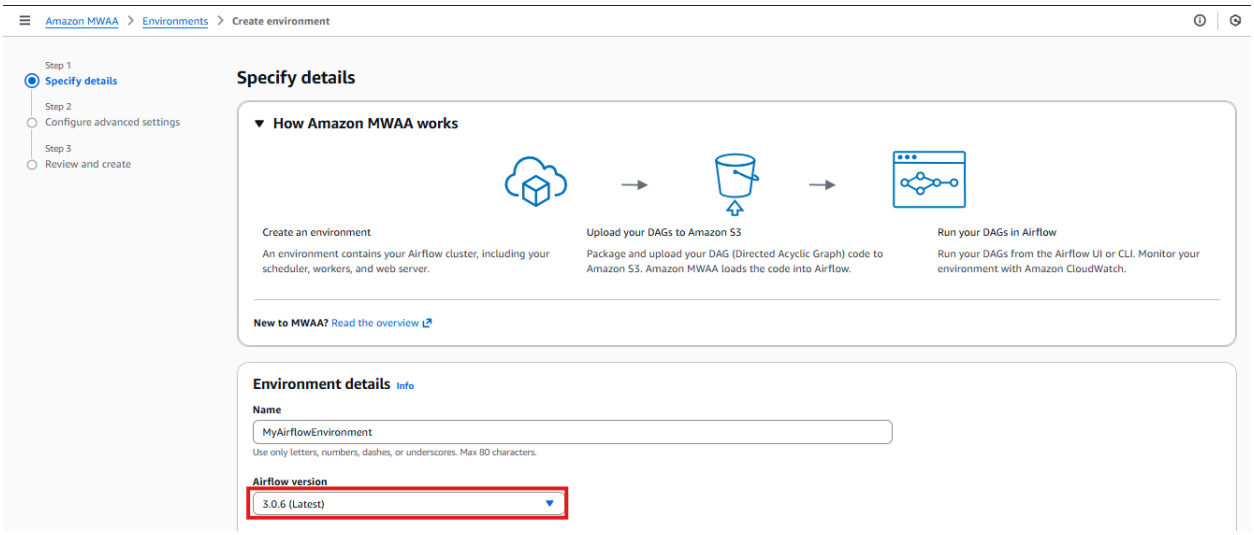


- Provide an environment name (e.g., workflows)



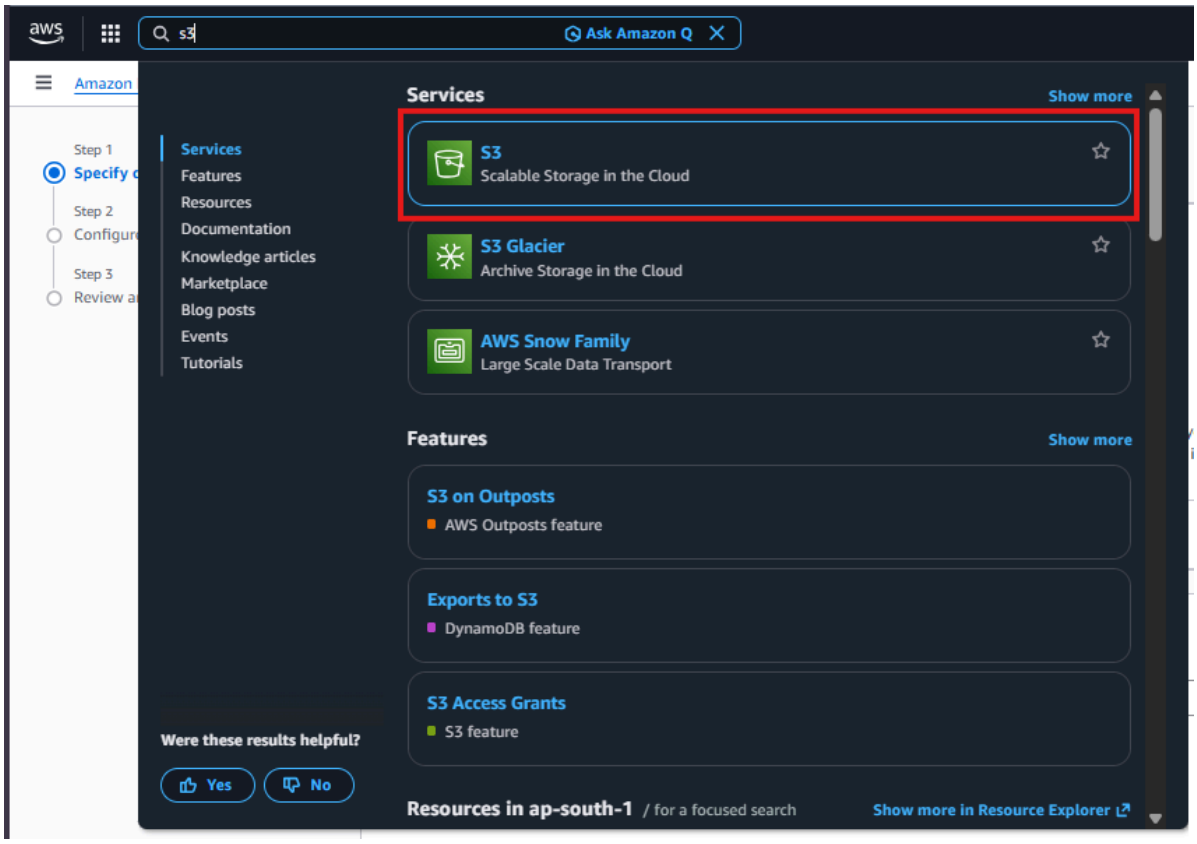


- Leave the Airflow version as the latest available version (e.g., 2.9.2)

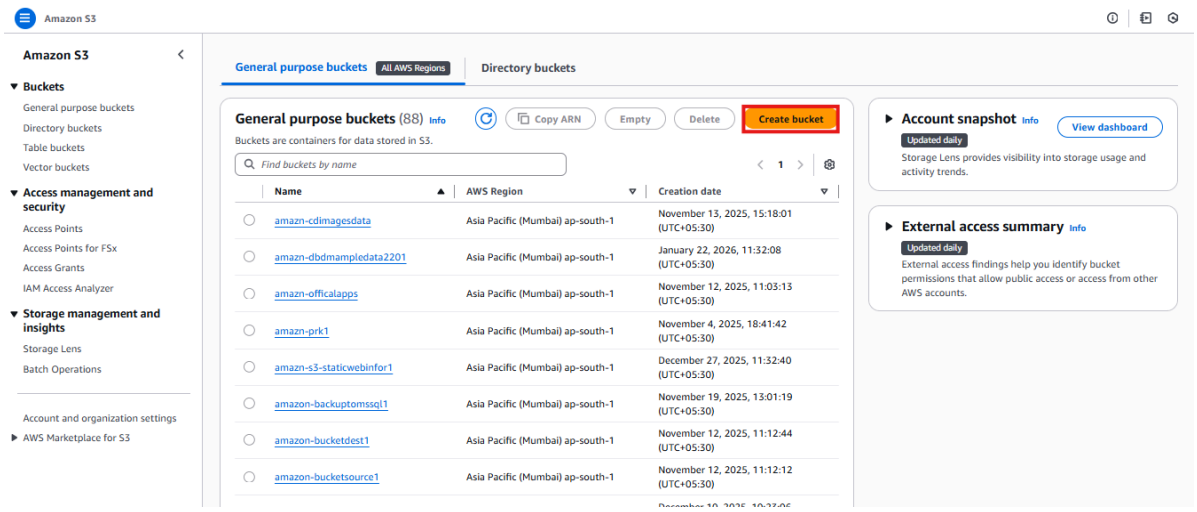


- Create or select an S3 bucket for Airflow ⟶ For creating follow the steps below:

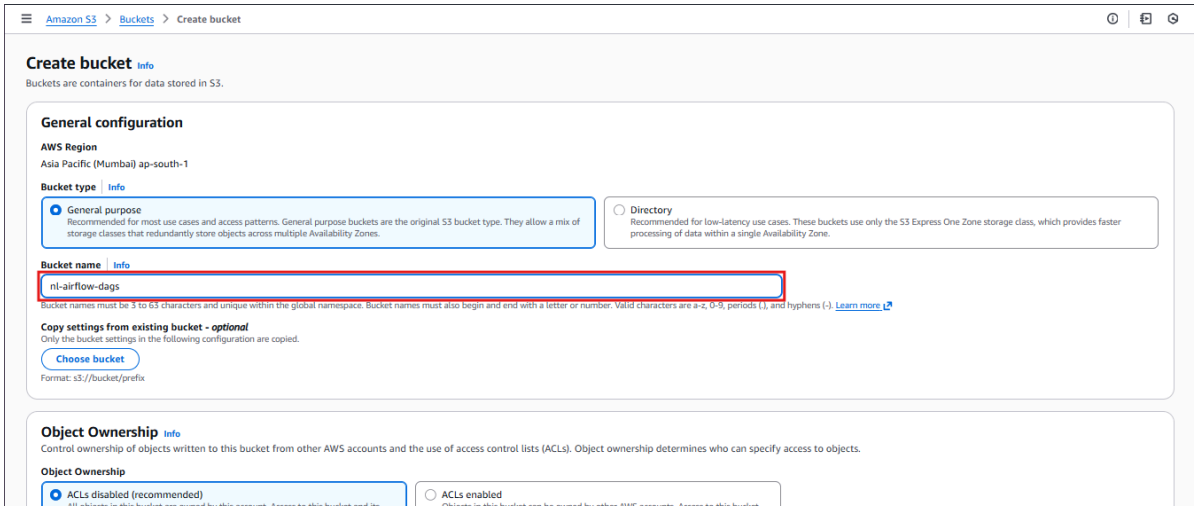
  1. Search S3 in the search bar —-→ Click on S3 (open in new tab)



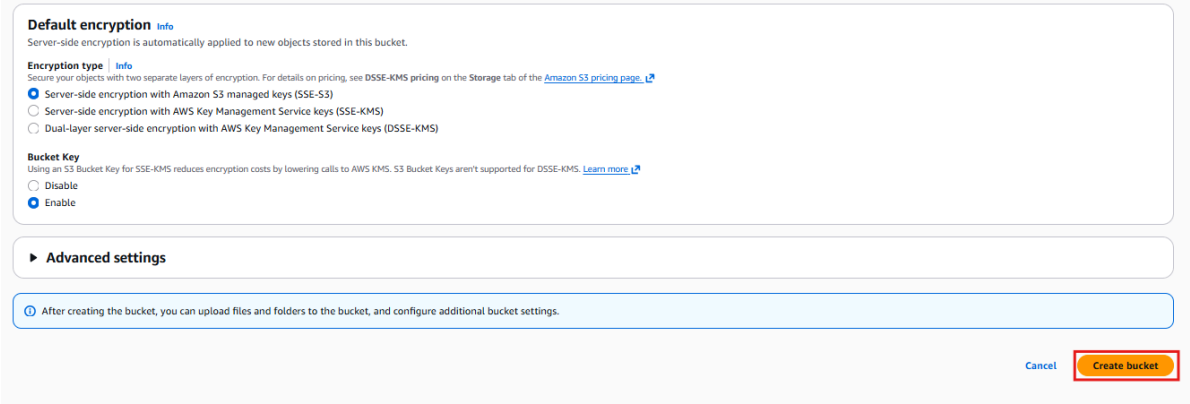
2. Click on **create bucket**



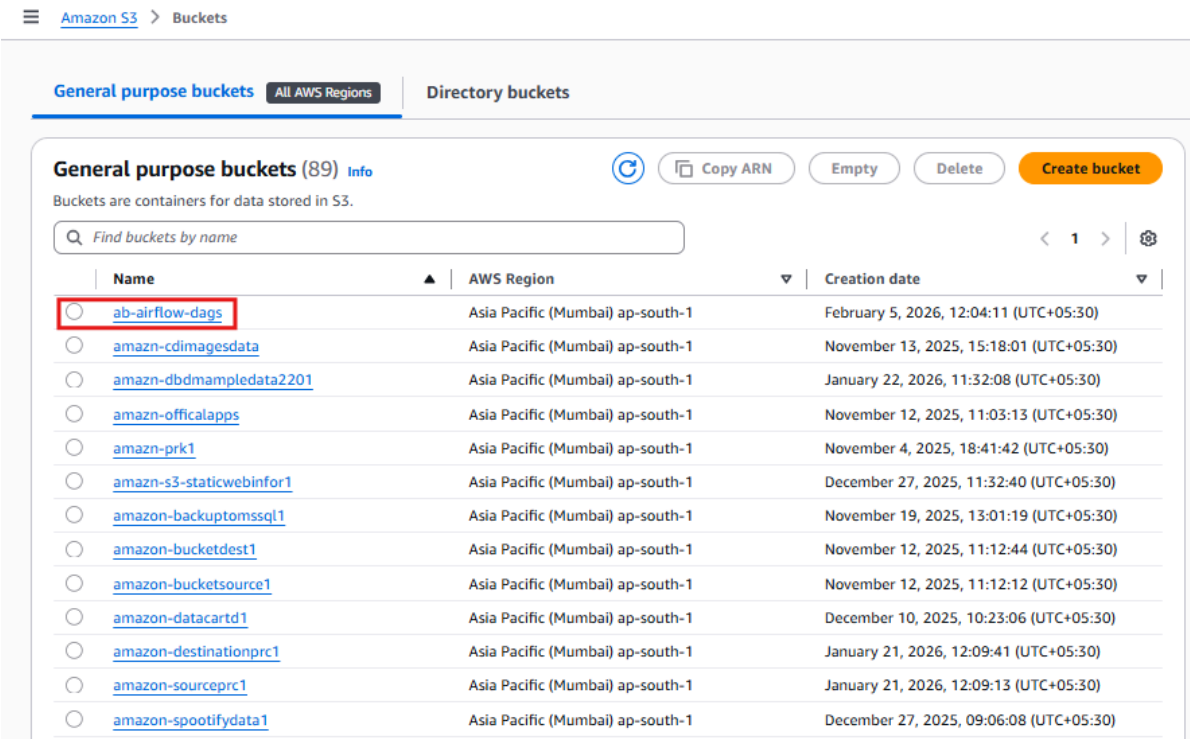
3. Give S3 bucket a **name** of your choice (eg ab-airflow-dags)



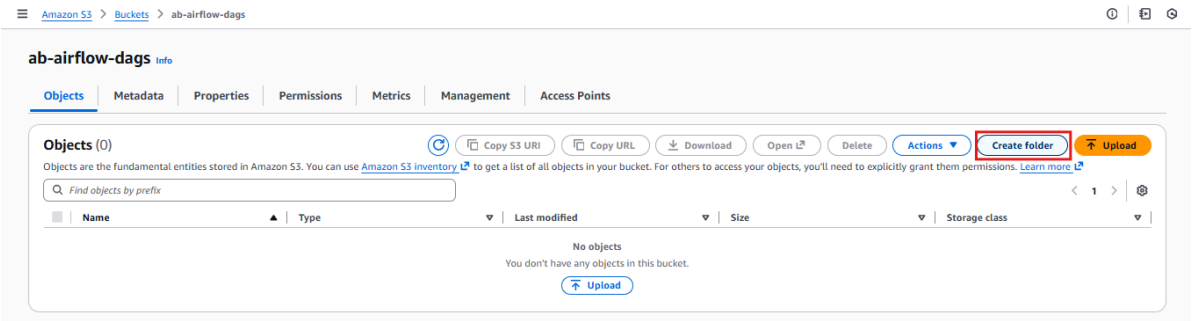
4. Scroll down and click on **Create bucket**

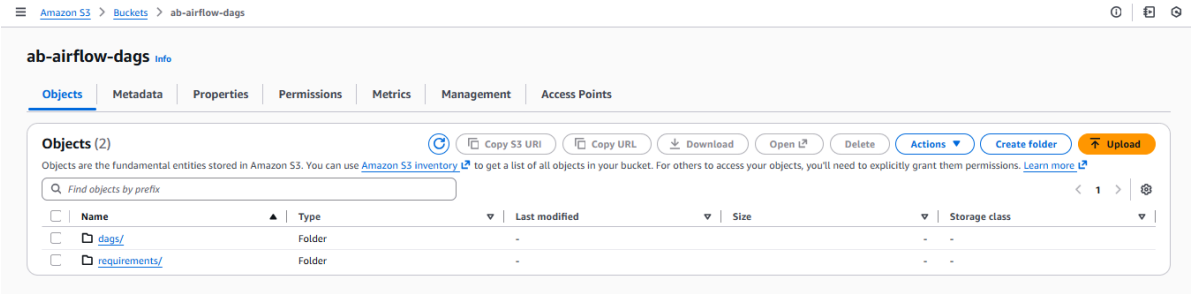


5. Click on newly created bucket

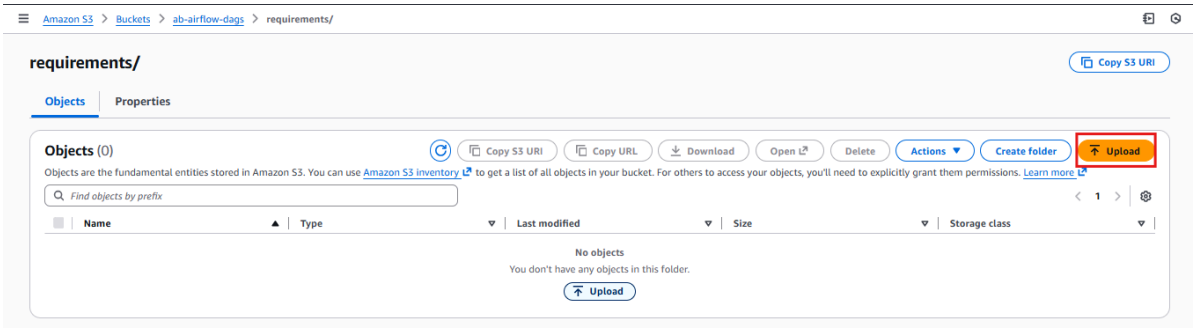


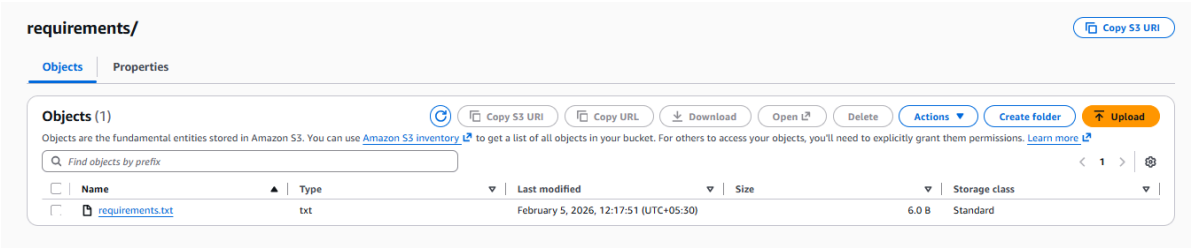
6. Inside the bucket, create two folders named `dags` and `requirements`



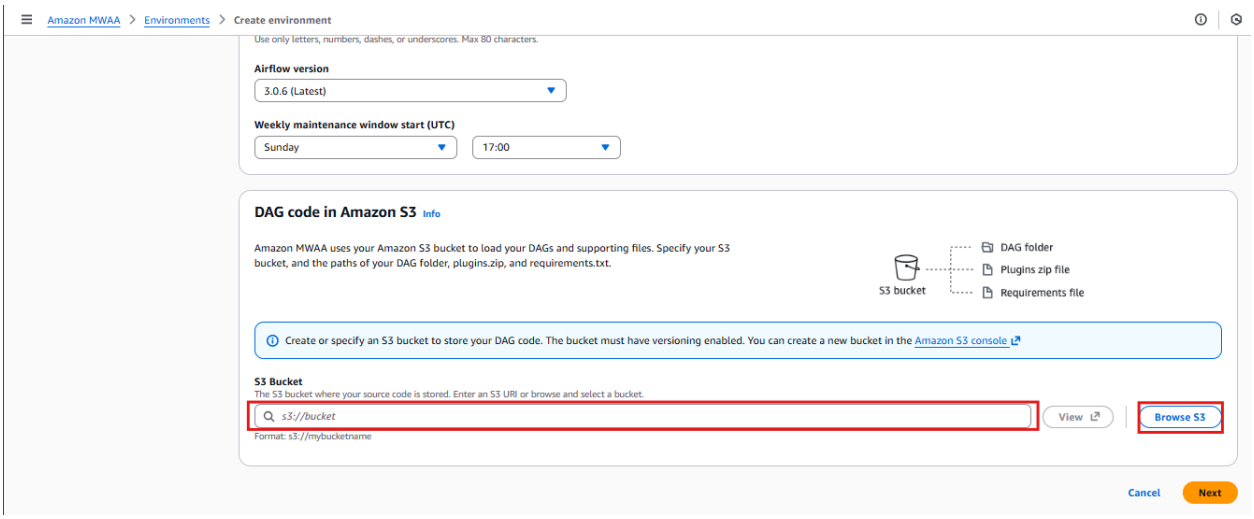


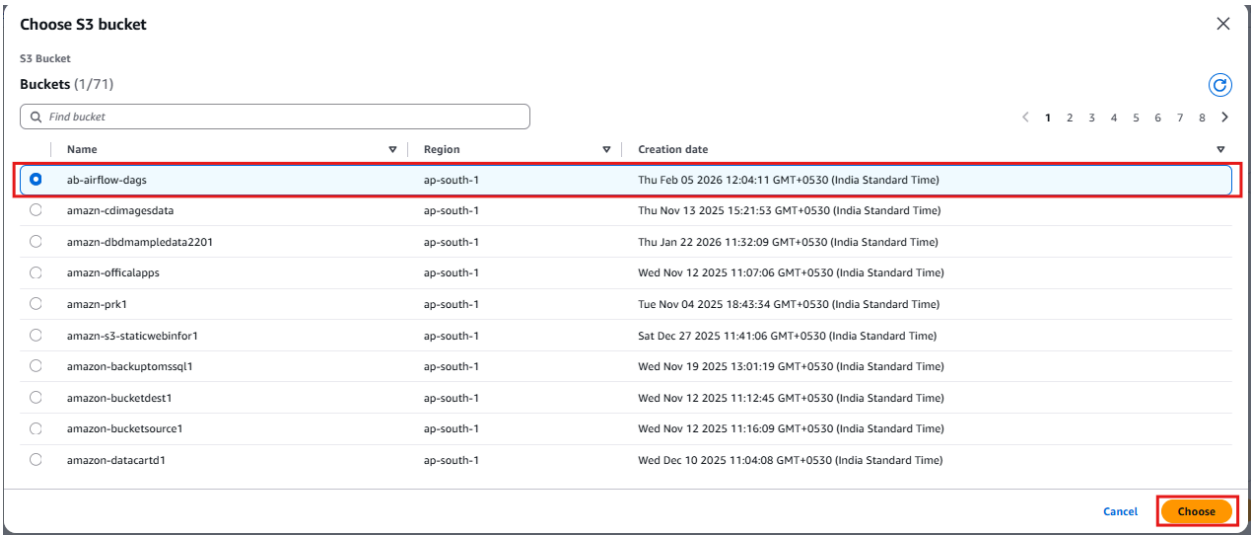
7. Upload a `requirements.txt` file inside the `requirements` folder

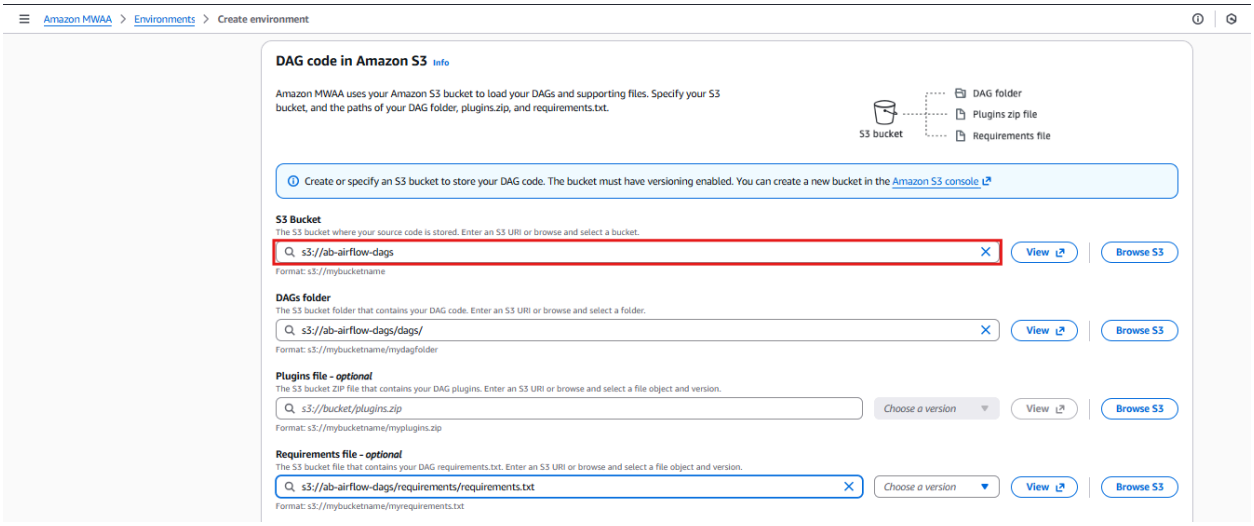




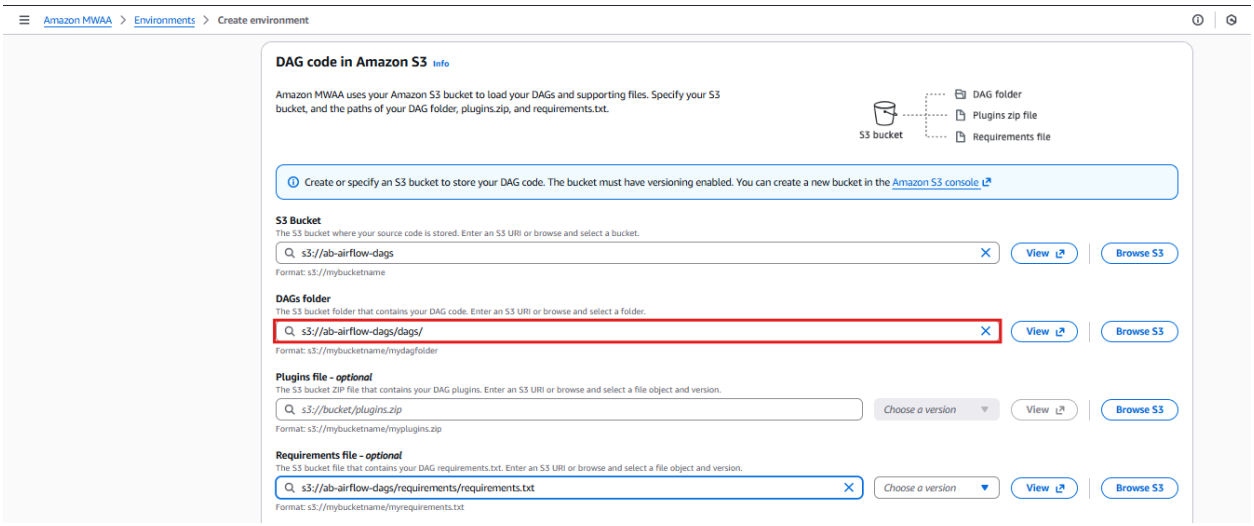
- Go back to Airflow ⟶ Click on browse to select the S3 bucket OR Enter the S3 bucket name in the DAGs configuration section


Amazon MWAA > Environments > Create environment
Use only letters, numbers, dashes, or underscores. Max 80 characters.
Airflow version
3.0.6 (Latest)
Weekly maintenance window start (UTC)
Sunday
17:00
DAG code in Amazon S3 Info
Amazon MWAA uses your Amazon S3 bucket to load your DAGs and supporting files. Specify your S3 bucket, and the paths of your DAG folder, plugins.zip, and requirements.txt.
S3 bucket
DAG folder
Plugins zip file
Requirements file
Create or specify an S3 bucket to store your DAG code. The bucket must have versioning enabled. You can create a new bucket in the Amazon S3 console
S3 Bucket
The S3 bucket where your source code is stored. Enter an S3 URI or browse and select a bucket.
s3://bucket
View
Browse S3
Format: s3://mybucketname
Cancel
Next

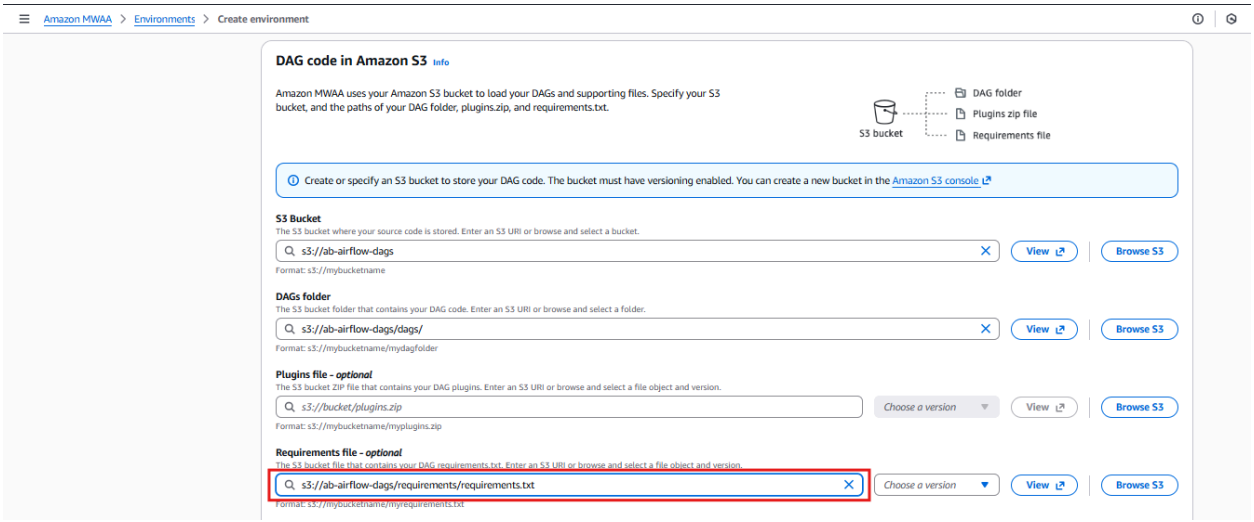

Choose S3 bucket
S3 Bucket
Buckets (1/71)
Find bucket
1 2 3 4 5 6 7 8
Name
Region
Creation date
ab-airflow-dags ap-south-1 Thu Feb 05 2026 12:04:11 GMT+0530 (India Standard Time)
amazn-cdimagesdata ap-south-1 Thu Nov 13 2025 15:21:53 GMT+0530 (India Standard Time)
amazn-dbdmampledata2201 ap-south-1 Thu Jan 22 2026 11:32:09 GMT+0530 (India Standard Time)
amazn-officalapps ap-south-1 Wed Nov 12 2025 11:07:06 GMT+0530 (India Standard Time)
amazn-prk1 ap-south-1 Tue Nov 04 2025 18:43:34 GMT+0530 (India Standard Time)
amazn-s3-staticwebinfor1 ap-south-1 Sat Dec 27 2025 11:41:06 GMT+0530 (India Standard Time)
amazon-backuptomssql1 ap-south-1 Wed Nov 19 2025 13:01:19 GMT+0530 (India Standard Time)
amazon-bucketdest1 ap-south-1 Wed Nov 12 2025 11:12:45 GMT+0530 (India Standard Time)
amazon-bucketsource1 ap-south-1 Wed Nov 12 2025 11:16:09 GMT+0530 (India Standard Time)
amazon-datacartd1 ap-south-1 Wed Dec 10 2025 11:04:08 GMT+0530 (India Standard Time)
Cancel
Choose


Amazon MWAA > Environments > Create environment
DAG code in Amazon S3 Info
Amazon MWAA uses your Amazon S3 bucket to load your DAGs and supporting files. Specify your S3 bucket, and the paths of your DAG folder, plugins.zip, and requirements.txt.
S3 bucket
DAG folder
Plugins zip file
Requirements file
Create or specify an S3 bucket to store your DAG code. The bucket must have versioning enabled. You can create a new bucket in the Amazon S3 console
S3 Bucket
The S3 bucket where your source code is stored. Enter an S3 URI or browse and select a bucket.
s3://ab-airflow-dags
View
Browse S3
Format: s3://mybucketname
DAGs folder
The S3 bucket folder that contains your DAG code. Enter an S3 URI or browse and select a folder.
s3://ab-airflow-dags/dags/
Format: s3://mybucketname/mydagfolder
Plugins file - optional
The S3 bucket ZIP file that contains your DAG plugins. Enter an S3 URI or browse and select a file object and version.
s3://bucket/plugins.zip
Choose a version
Format: s3://mybucketname/myplugins.zip
Requirements file - optional
The S3 bucket file that contains your DAG requirements.txt. Enter an S3 URI or browse and select a file object and version.
s3://ab-airflow-dags/requirements/requirements.txt
Format: s3://mybucketname/myrequirements.txt

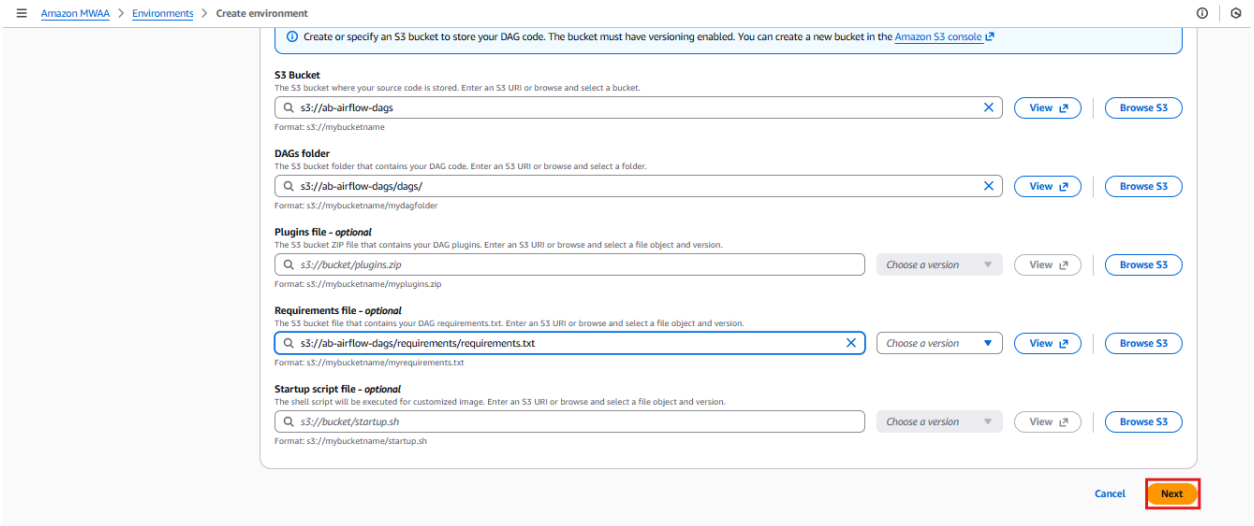
- Specify the DAGs folder path as `s3://<bucket-name>/dags`



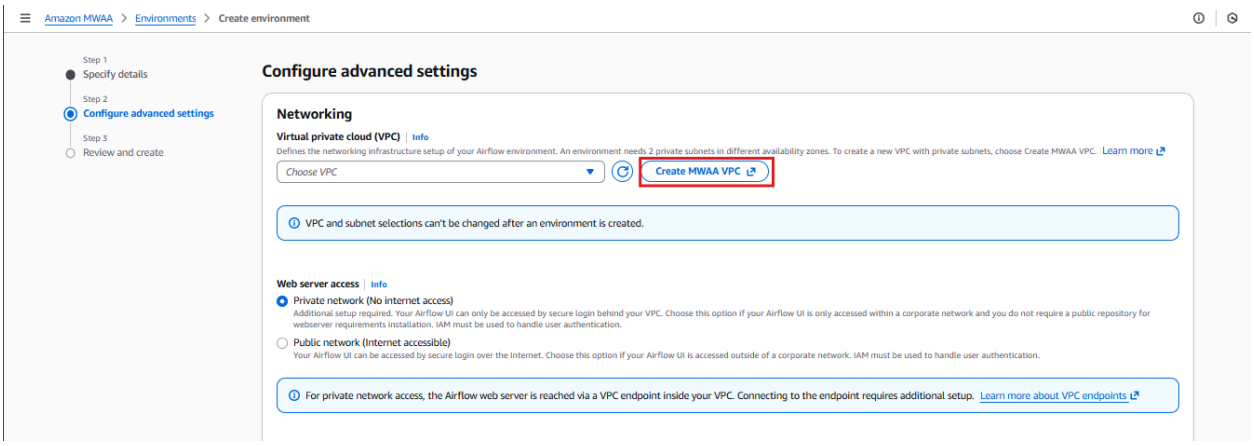
- Specify the requirements file path as `s3://<bucket-name>/requirements/requirements.txt`



- Click Next

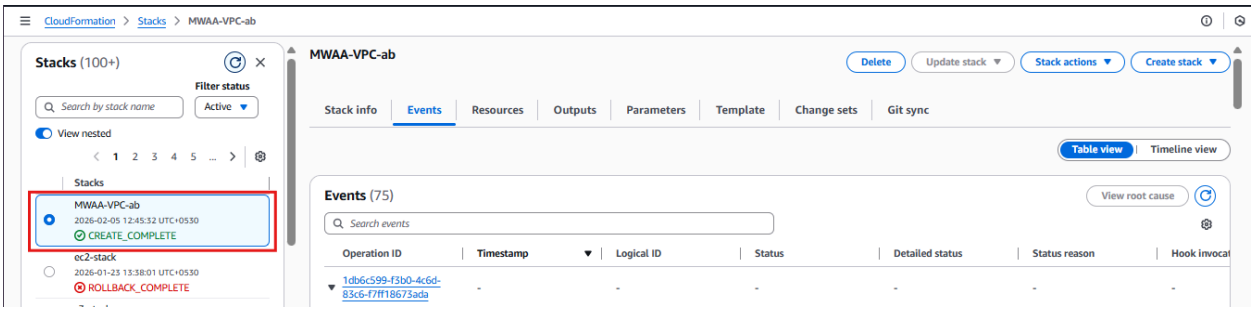


- Under Networking, click on Create MWAA VPC

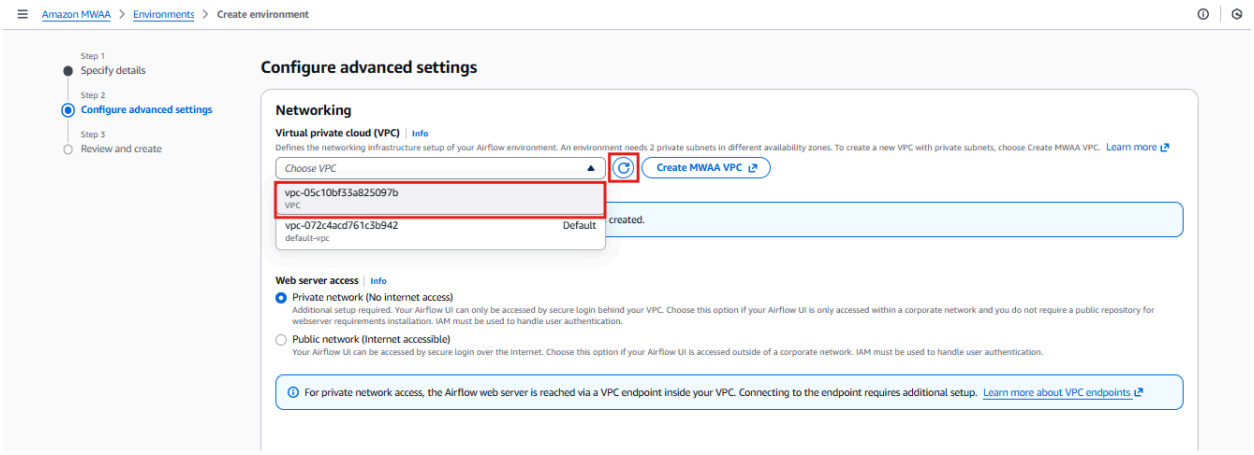


- Provide a stack name and leave all other values as default

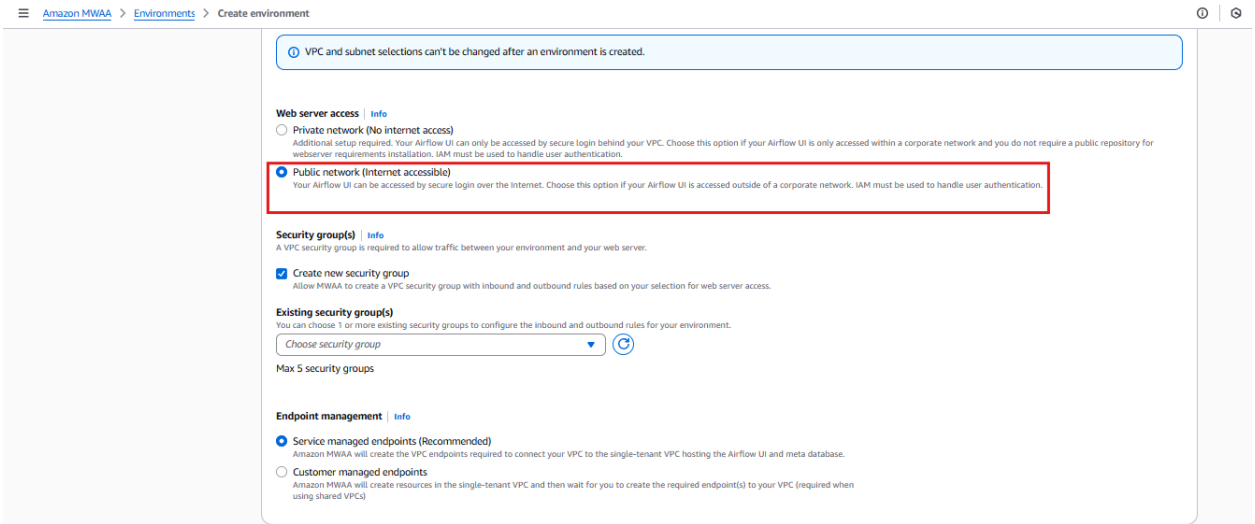
- Click Create stack and wait for the VPC creation to complete



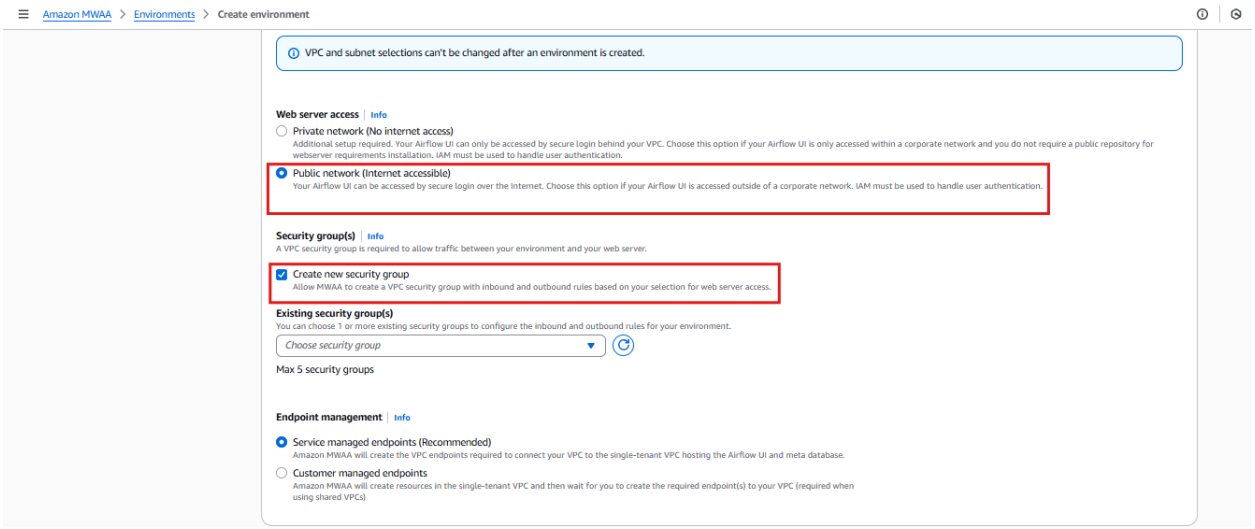
- Get back Airflow page ⟶ Refresh the VPC dropdown and select the newly created VPC



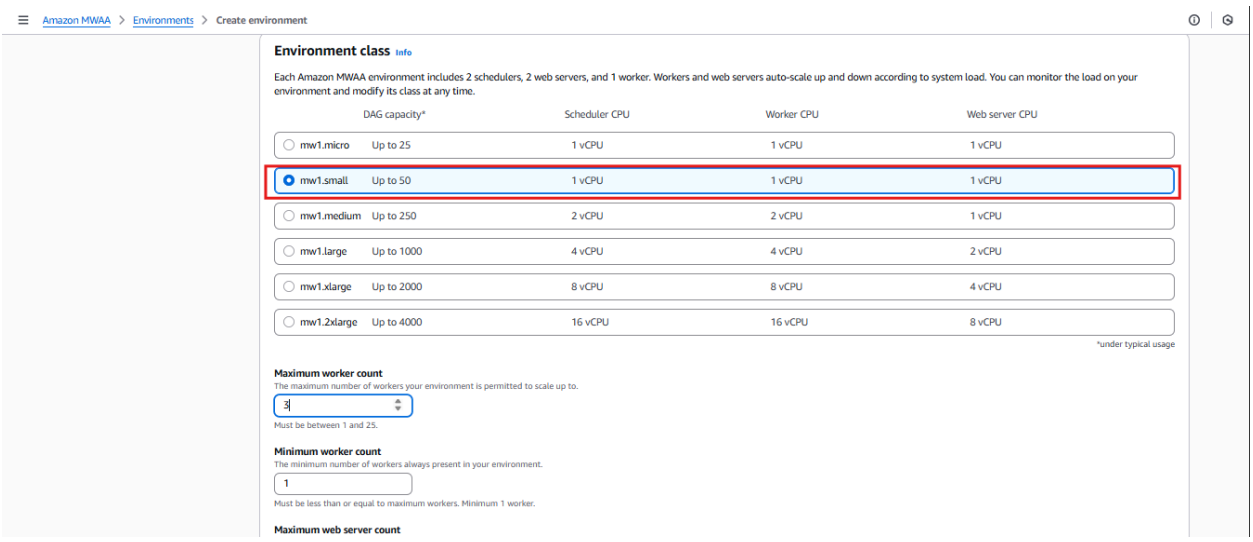
- Under Web server access, select Public network



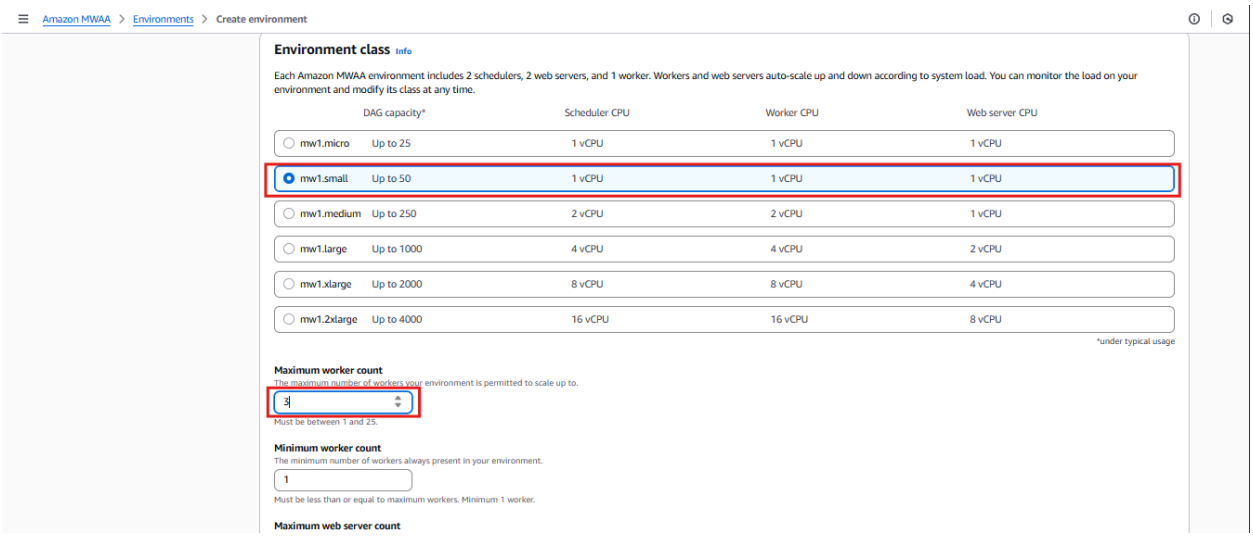
- Choose to create a new security group



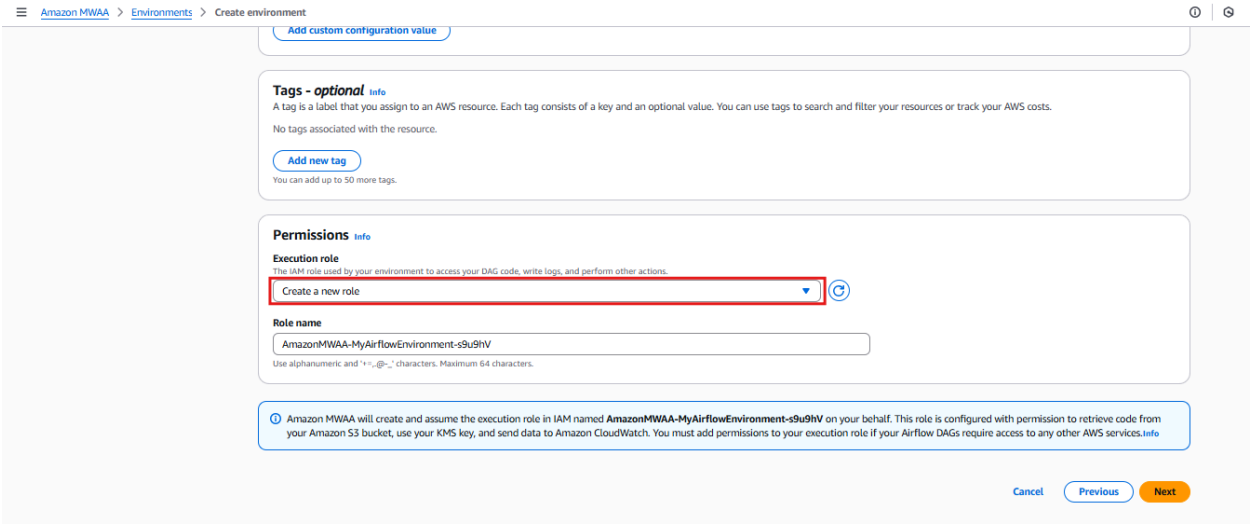
- Under Environment class, select Small



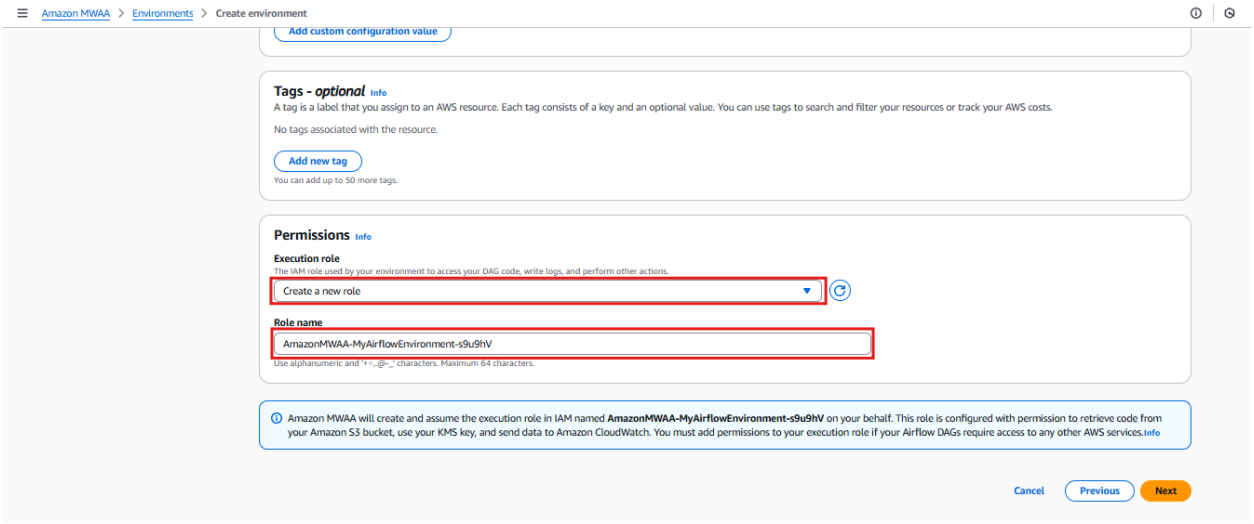
- Set the maximum worker count to 3



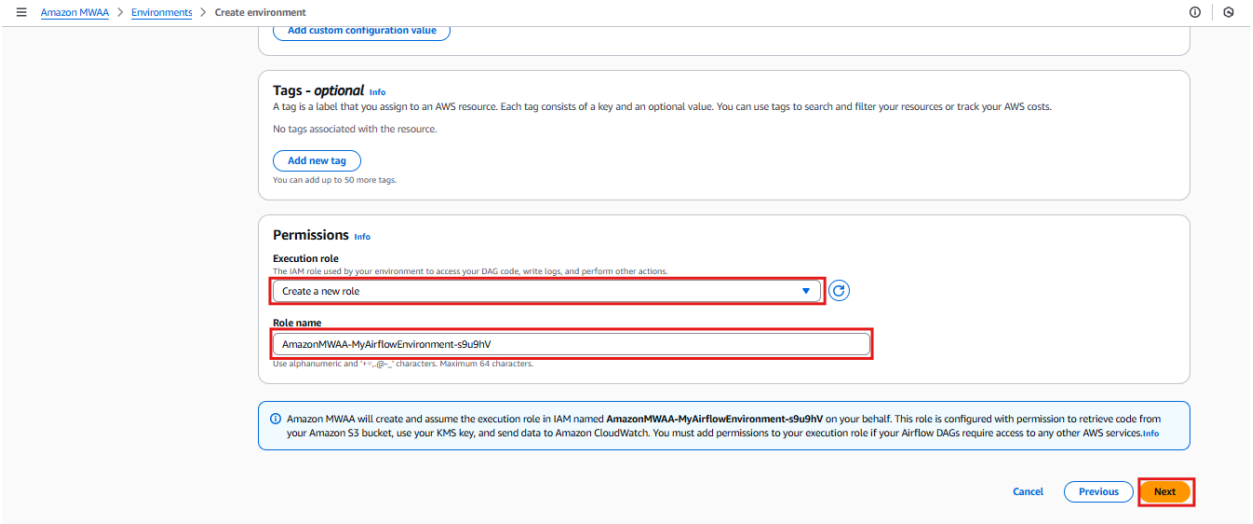
- Leave all other settings as default
- Under Execution role, select Create a new role

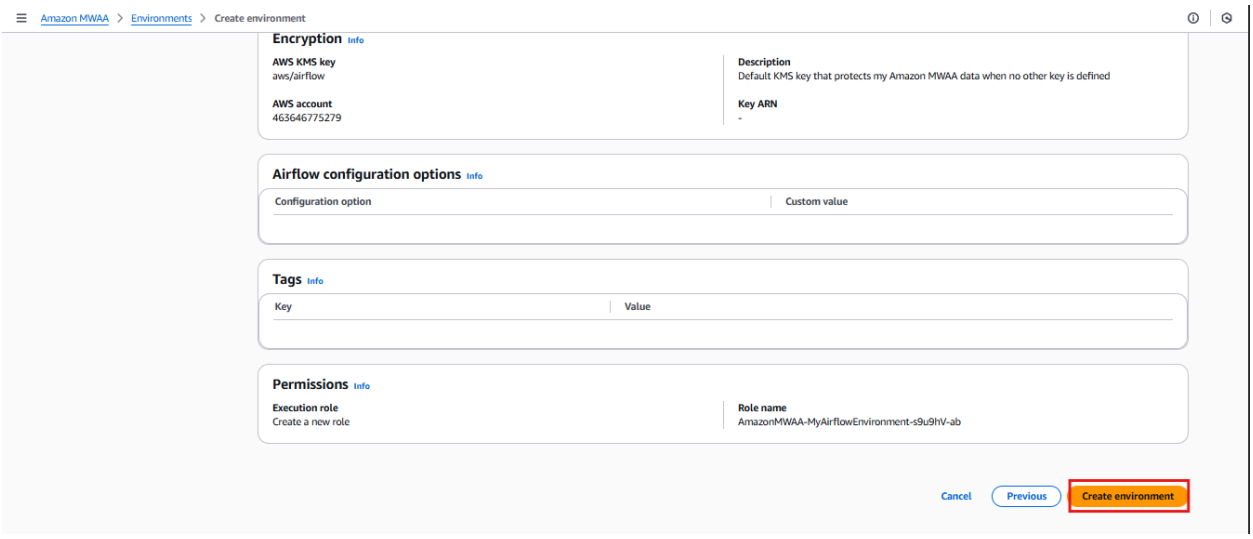


- Rename the role to a meaningful name (e.g., Airflow-Udemy-Role)

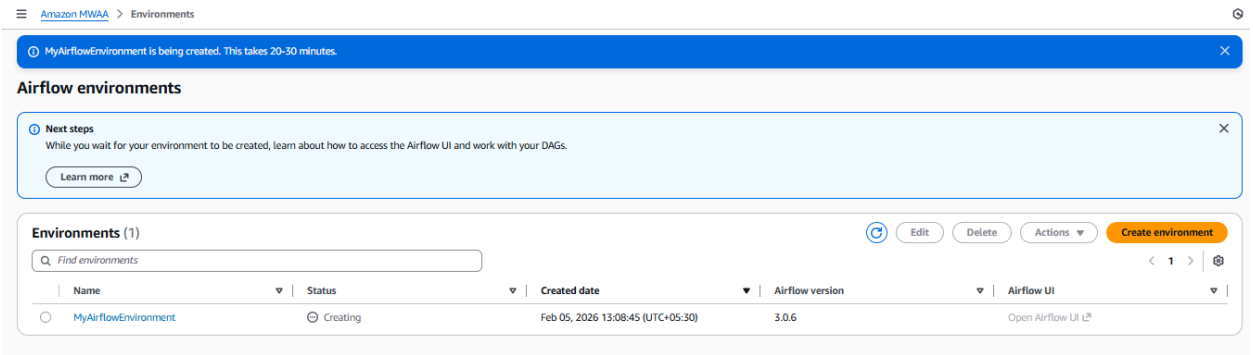


- Click Next, review the configuration, and click Create environment





- Wait for approximately 30 minutes for the environment creation to complete



Result

The AWS Managed Apache Airflow environment was successfully created with an S3 bucket for DAG storage, required dependencies configured through requirements.txt, a dedicated VPC, public web server access, and an execution IAM role. The Airflow environment is now ready for workflow orchestration.